

Problem

Determine $[\mathbf{y}]$ and $[\mathbf{T}]$ of a two-port network whose z parameters are $[\mathbf{z}] = \begin{bmatrix} 6 & 4 \\ 4 & 6 \end{bmatrix} \Omega$

Step-by-step solution

Step 1 of 5

Given Z parameters of two – port network is

$$[\mathbf{z}] = \begin{bmatrix} 6 & 4 \\ 4 & 6 \end{bmatrix} \Omega$$

If $z_{11} = 6, z_{12} = z_{21} = 4, z_{22} = 6$

Thus, the determinate of the matrix is

$$\Delta_z = z_{11}z_{22} - z_{12}z_{21}$$

$$\Delta_z = (6)(6) - (4)(4)$$

$$\Delta_z = 36 - 16$$

$$\Delta_z = 20$$

Step 2 of 5

Admittance parameters

$$y_{11} = \frac{z_{22}}{\Delta_z}$$

$$y_{11} = \frac{6}{20}$$

$$\therefore y_{11} = 0.3 \text{ S}$$

$$y_{12} = \frac{-z_{12}}{\Delta_z}$$

$$y_{12} = \frac{-4}{20}$$

$$\therefore y_{12} = -0.2 \text{ S}$$

Step 3 of 5

$$y_{21} = -\frac{z_{21}}{\Delta_z}$$

$$y_{21} = \frac{-4}{20}$$

$$\therefore y_{21} = -0.2S$$

$$y_{22} = \frac{z_{11}}{\Delta_z}$$

$$y_{22} = \frac{6}{20}$$

$$\therefore y_{22} = 0.3S$$

Thus, the y – parameters are

$$[y] = \begin{bmatrix} 0.3 & -0.2 \\ -0.2 & 0.3 \end{bmatrix}$$

Step 4 of 5

Transmission parameters

$$[T] = \begin{bmatrix} A & B \\ C & D \end{bmatrix}$$

$$A = \frac{z_{11}}{z_{21}}$$

$$A = \frac{6}{4}$$

$$\therefore A = 1.5$$

Step 5 of 5

$$B = \frac{\Delta_z}{z_{21}}$$

$$B = \frac{20}{4}$$

$$\therefore B = 5\Omega$$

$$C = \frac{1}{z_{21}}$$

$$C = \frac{1}{4}$$

$$\therefore C = 0.25S$$

$$D = \frac{z_{22}}{z_{21}}$$

$$D = \frac{6}{4}$$

$$\therefore D = 1.5$$

Thus, the T – parameter

$$[T] = \begin{bmatrix} 1.5 & 5\Omega \\ 0.25S & 1.5 \end{bmatrix}$$